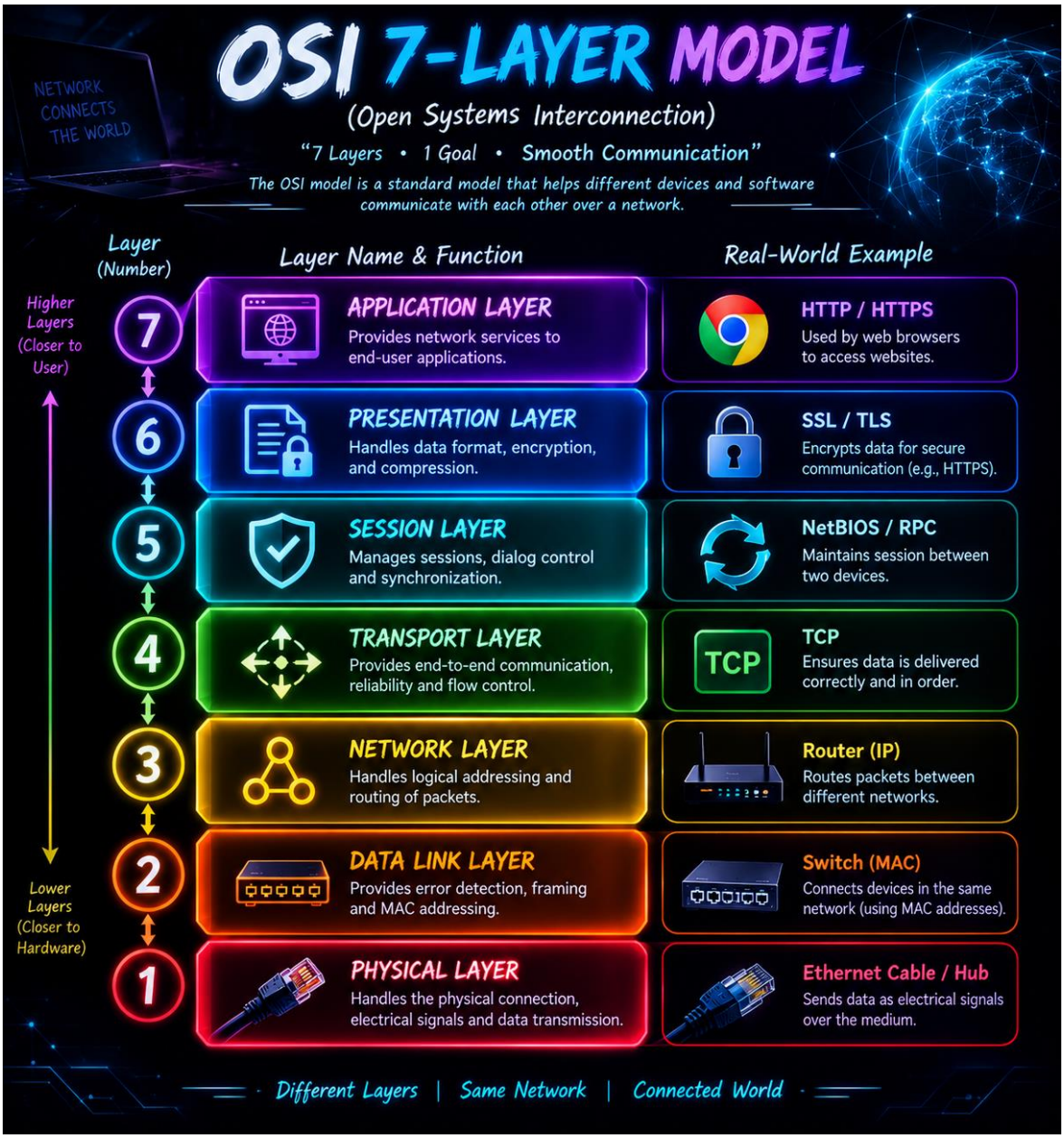
1 Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).



2 . Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

No.	Scenario	Correct OSI Layer
a)	Encrypting data before sending	Presentation Layer (Layer 6)
b)	Routing packets across networks	Network Layer (Layer 3)
c)	Converting data to electrical signals	Physical Layer (Layer 1)
d)	Opening a web page in your browser	Application Layer (Layer 7)
e)	Detecting and correcting errors in frames	Data Link Layer (Layer 2)

3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

App name : What App

When I send or receive a message on WhatsApp, different OSI layers work together to transfer data between devices.

1. Application Layer (Layer 7): WhatsApp allows me to send and receive messages, photos, and videos.

2. Presentation Layer (Layer 6): Data is encrypted to help protect messages during communication.

3. Transport Layer (Layer 4): It helps transfer data between devices reliably using transport protocols.

4. Network Layer (Layer 3): IP addresses help route data from my phone to the destination server and back.

5. Data Link Layer (Layer 2): Wi-Fi or Ethernet carries data in frames across the local network.

6. Physical Layer (Layer 1): Wi-Fi radio waves or electrical signals carry the data through the network medium.

Conclusion: These OSI layers work together to help WhatsApp send and receive messages over a network.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

Website Not Opening but Ping Is Successful

1. Application Layer (Layer 7): The problem may be in the web browser, website server, or HTTP/HTTPS service.

2. Presentation Layer (Layer 6): There may be an SSL/TLS certificate or encryption problem that prevents a secure connection.

3. Session Layer (Layer 5): There may be a problem establishing or maintaining the communication session.

Reasoning: Since we can successfully ping the website's IP address, basic IP connectivity is working. However, the website may still be inaccessible because of a browser issue, DNS problem, firewall rule, or HTTP/HTTPS service failure.

Conclusion: The Application Layer (Layer 7) is a likely place to investigate first, although the problem could also involve other layers or services.